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Impact of Minimum Wage Policy on Youth Employment in the United States

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Introduction

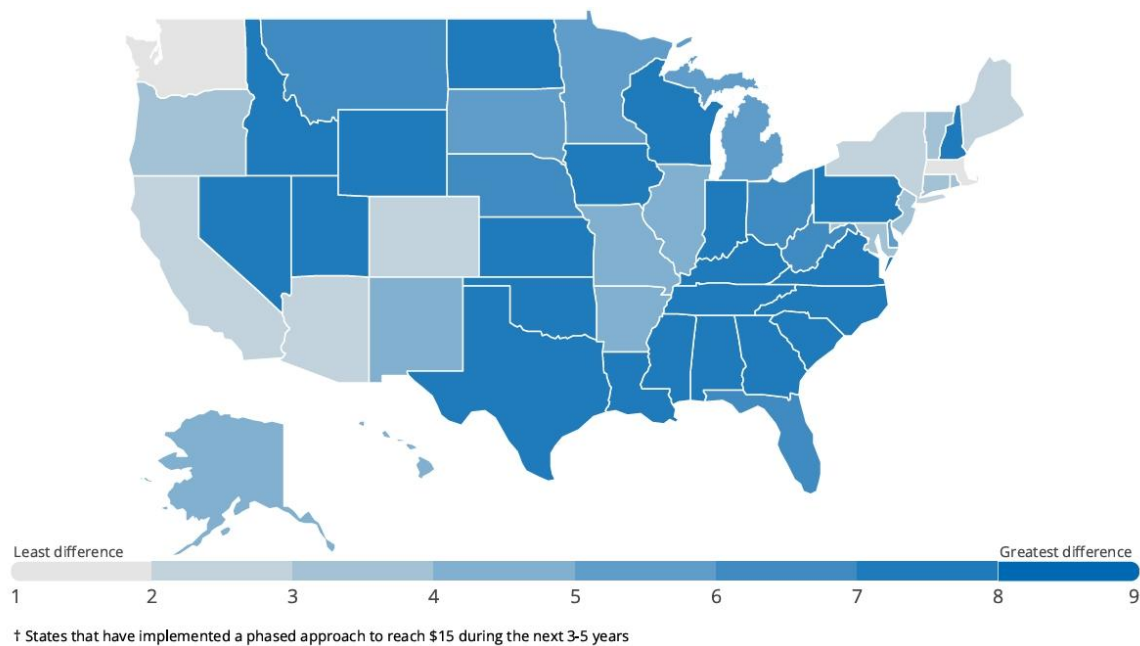
The employment impact of statutory wage floors is one of the most contested issues in labour economics. Teenagers aged 16-19 are central to the debate as they cluster in low-wage sectors with higher unemployment rates than mature workers. The theoretical expectation under a competitive labour market is that imposing a wage floor above the market equilibrium will reduce employment, particularly for less experienced workers; however, empirical findings are yet to settle on the true magnitude of this effect (Meer & West, 2016). As of 2025, the US stands out for having the lowest unchanged federal wage floor across the OECD, remaining at \$7.25 per hour, last revised in 2009. While state-level policy has partly offset this stagnation, the US Katz index (minimum to median wage ratio) remains at approximately 45%, well below the OECD average of 56.7% (OECD, 2025).

The US Labour market outcomes for teenagers vary drastically by gender, ethnicity and age. In March 2025, the employment-to-population ratio of females aged 16-19 (33.1%) exceeded that of males (31.7%), yet both groups faced high unemployment rates of 13.3% and 14.2%, respectively (U.S. Bureau of Labour Statistics, 2025). Racial disparities persist as White, Hispanic and Black teenagers had unemployment rates of 12.7%, 16% and 20.8% respectively (BLS, 2025a). The importance of early work opportunities further drives research relevance in this field. Youth employment is viewed as a pathway to skill development and effective signalling for future employment. Workers under 25 comprise roughly 44% of the labour workforce paid at or below minimum wage in the US (BLS, 2024b), making them particularly sensitive to policy changes in wage floors. Numerous states have introduced substantially higher rates to combat high costs of living and wage indexing to adjust for inflation; these policy shifts provide natural experiments to study the impact of minimum wage policy on the 16-19 age group. Therefore, on a country-level comparison, it is essential to ask whether a higher minimum wage reduces teenage employment on average and what the cohort-specific effects and policy responses are.

Background

The United States federal minimum wage was established in 1938 and has been raised intermittently till July of 2009. The prolonged stagnation has caused 30 states and many cities to set higher minimum wages and mandate further steady increases, as seen in Figure 1.

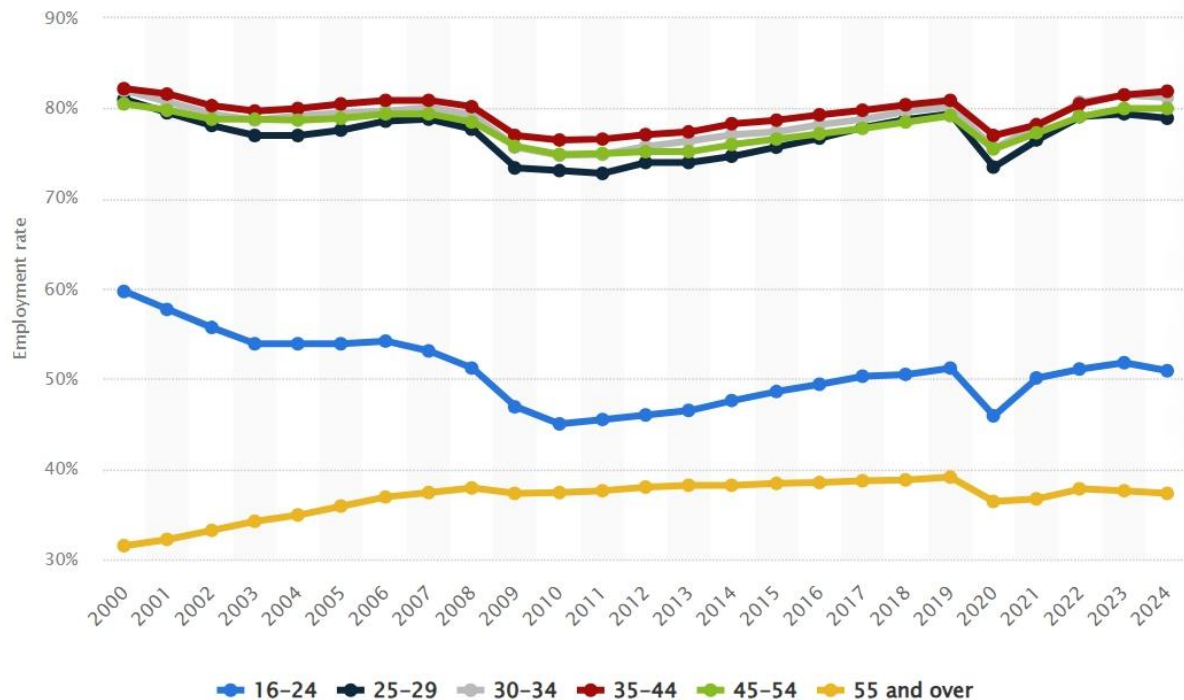
Figure 1: Proposed \$15 state minimum wage



Source: Brantley, 2021

Proponents of higher minimum wage argue that they boost earnings for low-income workers and reduce inequality, while opponents argue that a higher wage floor will induce the substitution effect, resulting in lower employment of teenagers (Manning, 2021). The US labour law tries to mitigate this by the Fair Labour Standards Act introduced in 1996 (Mayer et. al, 2013), allowing employers to pay certain categories of youth workers \$4.25 for the first 90 days of employment, encouraging the hiring process. Such exceptions are narrow, and most working teens are subject to the prevailing federal or state minimum wage. As illustrated in Figure 2, the employment rate for youth workers 16–24 has consistently remained lower than that of mature workers.

Figure 2: Employment By Age



Source: Statista

Inferring from this data, it is crucial to consider the role of education in youth labour supply. Individuals in school typically opt for part-time work, while those out of school seek full-time jobs, as shown in (BLS, 2024c). The effects of minimum wage can differ between these groups, potentially deterring employers from hiring students or limiting job access for those no longer in education (Geel & Backes, 2012). Some countries, like the UK, address this issue with age-specific minimum wages to preserve youth employment. The following tables outline key changes in the U.S. federal minimum wage since 1990, and a few state-specific wage rises highlighting the policy intuition behind each of them.

Table 1: US Federal Minimum Wage Timeline

Effective Date	Wage	Policy
April 1, 1990	\$3.80	Addressed the fall of real wages due to inflation after a 9-year stagnation period to restore purchasing power for low-wage workers.
April 1, 1991	\$4.25	1989 Fair Labour Standards Amendments: continued effort to improve earnings at the bottom of the wage distribution.
October 1, 1996	\$4.75	Part of the Small Business Job Protection Act aimed to reduce income inequality while providing tax offsets to small businesses.
September 1, 1997	\$5.15	Final stage of the 1996 increase; intended to further raise low-end wages while minimising employment effects.
July 24, 2007	\$5.85	Enacted under the Fair Minimum Wage Act of 2007 in response to declining real wages and rising living costs.
July 24, 2008	\$6.55	Continued effect of the 2007 Act; to reduce income inequality among the working class amid economic stagnation.
July 24, 2009	\$7.25	Final stage of the 2007 Act; implemented during the Great Recession to stimulate aggregate demand.

Source: UDL 2019

Table 2: US Current State Minimum Wage (Highest wages)

State	Wage	Policy
Connecticut	\$16.35	The 2019 law linking wage increases to the federal Employment Cost Index.
California	\$16.50	State legislation mandates annual increases, adjusted for inflation.
New York	\$16.50 (NYC & select counties)	Regional wages for NYC and surrounding counties are set at higher rates.
New Jersey	\$15.49	State law with annual increases, adjusted for inflation.
Washington	\$16.6	Annual inflation adjustment via CPI, as a voter-approved initiative.

Source: BLS 2024d

Literature Review

Theoretical Expectations

The difference between state and federal minimum wages requires an economic explanation relying on the institutional and dynamic monopsony models. In perfect competition, firms adjust by raising prices and cutting non-wage benefits to preserve employment (Schmitt, 2013). Labour institutions see a fall in real wages following a wage rise, while a rising minimum wage encourages firms to enhance efficiency and productivity. Data from the UK after the 1996 Employment Rights Act supports this theory, as UK-based automobile firms ranked among the most productive internationally (Freeman, 1998). In agreement, Nickell & Layard (1999) argue that trade unions and social security systems influence unemployment and growth more than minimum wage changes. Their research shows that coordinated bargaining networks, as seen in Sweden and Austria, show rising employment due to coordinated union networks. Within these systems, wage-to-effort bargaining forms a segmented labour market across private and public sectors, where wages reflect a strategic equilibrium balancing cooperation and opportunism (Kaufman, 2012).

The empirical research on youth labour unemployment is extensive. Macroeconomic factors, particularly adult unemployment, have confounding effects on youth unemployment. Blanchflower & Freeman (2000) find a strong positive correlation between adult and youth unemployment rates, with a 1% rise in adult unemployment leading to a 0.73% increase in youth unemployment. With an established relationship to broader macroeconomic phenomena, O'Higgins (2015) notes the importance of sustained economic growth for increasing youth employment. The study also categorises individuals into NEET (not in employment, education, or training), which provides a more accurate measure of the labour market, representing discouraged individuals who have been out of the labour market for qualification-related reasons.

An important finding that supports the market segmentation theory is the negative impact of flexible work contracts on youth employees. Workers are trapped in precarious short-term or part-time employment, making it challenging to transition to secure jobs. In contrast, Clark & Summers (1982) support the aggregate demand theory, contending that the core problem is a shortage of jobs, not a lack of skilled workers. The study shows that teenage unemployment responds strongly to labour market conditions; a reduction in unemployment rates for skilled adults by one percentage point increases employment for teenagers by 4%. Between 1969 and

1976, the aggregate unemployment rate rose from 3.6% to 7.7%, and the proportion of unemployed teens increased fourfold, with a sixfold higher unemployment rate for black teenagers. These figures represent the long-term unemployment rates, essentially indicating that teenagers move between job sectors with little difficulty.

The dynamic monopsony model addresses the relationship between firms and minimum wage changes. Many labour markets operate under monopsony, where employers set wages as workers face uncertainty when switching jobs. Asher et al. (2010), building on Manning (2003), find that even in competitive markets, imperfect information and job differentiation provide firms with wage-setting advantage. Although classical theory suggests firms must raise wages to attract labour, financial and geographic barriers allow firms to fill low-wage roles without significant pay rises. Therefore, raising the minimum wage can benefit new labour market entrants. US data from the 1970s estimates minor adverse teen employment effects between 0.1 and 0.3. However, more recent studies find no consistent impact, with heterogeneity across workers and firms still allowing for some job losses (Boal & Ransom, 1997). A meta-analysis of 53 studies finds lower elasticities in public sector jobs like nursing and teaching, indicating a stronger presence of monopsony, Sokolova and Sorensen (2021) conclude that minimum wage increases are unlikely to cause widespread job losses under monopsonistic conditions.

Empirical Evidence

The most notable study to utilise a natural experiment and contribute to the literature on minimum wage impacts on low-income jobs was Card & Krueger's (1994) analysis of the New Jersey wage rise on employment in the fast-food industry. The research used Pennsylvania as the control group to employ the Differences in Differences (DID) methodology and found no job loss among teenagers due to higher wages and no reduction in non-wage benefits, despite a slight increase in fast food prices. To generalise these findings, Dube et al. (2010) transpose the DID model to the national level and reduce bias by comparing border counties, as they are economically similar, to account for state-level heterogeneity.

Zavodny (2000) studies the impact of minimum wage increases on U.S. teen employment and working hours between 1979 and 1993. Utilising both state-level and individual-level data, the research finds that a 10% rise in the minimum wage results in a slight reduction in teen employment by 1.2%, while having no significant effect on weekly hours worked. Ragan (1977) uses national time-series regressions on quarterly data from 1963 to 1972 to study

minimum wage effects on youth labour markets. The study finds that minimum wage increases reduced youth employment by 3.3% and raised unemployment by 3.8 percentage points.

Contemporary studies on minimum wage focus on the distribution of wage impacts among lower-wage earners, particularly teenagers, as data suggests. Traditional DID approaches measure the average employment effects, which leads to a loss of observable heterogeneity across groups. To address these issues, Biewen et al. (2022) use distribution regression DID to estimate the probability of a worker's wage falling below certain thresholds. The findings show wages below €8.50 were almost eliminated, resulting in a reduction of wage inequality. In contrast, no significant changes were observed in the distribution of weekly working hours, indicating that firms adjusted wages rather than cutting hours. To study the impact of time-varying effects on wage rises, Cengiz et al. (2019) employ dynamic time trend-specific DID to find the long-term effects of minimum wage shocks and identify no significant job losses five years after the wage rise, suggesting that the minimum wage increase redistributes jobs rather than removes them.

Data collected

The Current Population Survey is a national monthly survey of labour market variations in the U.S. The March dataset is utilised to maintain consistency and ensure comparable results for longitudinal data, as policy shocks are time-sensitive. This study focuses on the annual employment status outcomes collected and aggregates individual responses sorted into yearly state cells, specifically for teenagers aged 16 to 19 who were employed in each state at various points in time. Additionally, we include the Consumer Price Index (CPI) for March (Federal Reserve Bank of St. Louis, 2024), the nominal wage in effect in March for each state (*Neumark et. al., 2014*) and the federal appointed minimum wage in March of each year (*U.S. Department of Labor, 2009*). The nominal minimum wage at the state and federal level is adjusted for inflation using CPI to create the real minimum wage for time variant consistency. Employment fractions are noted between 1 and 0, and wages are recorded in USD. Working status distinguishes the sector of employment and contract hours, and the gender and ethnic composition of those groups.

Table 3: Variable definitions

Variable	Definition
employed_percent	Fraction of teens (aged 16–19) employed in March, as a percentage.
employedf_percent	Fraction of employed female teens, as a percentage.
employedm_percent	Fraction of employed male teens, expressed as a percentage.
work_fulltime_percent	Fraction of teens employed full-time (≥ 30 hours/week), as a percentage.
work_parttime_percent	Fraction of teens employed part-time (< 30 hours/week), as a percentage.
work_parttime_female_percent	Fraction of females working part-time, percentage.
work_parttime_male_percent	Fraction of males working part-time, percentage.
work_fulltime_female_percent	Fraction of females working full-time, percentage
work_fulltime_male_percent	Fraction of males working full-time, percentage.
minwage	The nominal state minimum wage that was in effect in March.
fed_minwage	The federal minimum wage that was in effect in March.
cpi	Consumer Price Index (March, base).
real_minwage	State minimum wage adjusted for inflation ($\text{minwage} \div \text{CPI}$).
ln_real_minwage	Natural logarithm of real minimum wage.
state_above_fed	Dummy = 1 if state minwage > federal minwage by > \$0.009 and 0.005 for real minimum wage
minwage_difference	Difference between state and federal minimum wages
year_dummy	Dummy = 1 in the first year a state exceeds the federal minimum wage
treat_year	The first year when a state's minimum wage exceeds the federal minimum wage and the subsequent years
treatment	Dummy = 1 if state treated (minimum wage > federal)
post	Dummy = 1 if year is post-treatment for treated states
treatment_post	Interaction of treatment and post-dummies (DID)

Table 4: Summary statistics of key variables

Variables	(1) No. of observations	(2) Mean	(3) Standard Deviation	(4) Minimum	(5) Maximum	(6) Skewness	(7) Kurtosis
Employment percentage	1224	36.497	10.112	4.912	68.033	.055	2.932
Part-time employment percentage	1224	25.368	8.193	2.738	53.121	.307	2.993
Full-time employment percentage	1224	11.13	4.406	1.244	33.381	.556	3.477
Percentage of teenage employees who are in school	1224	80.875	5.41	60.707	94.591	-.456	3.216
Percentage of teenagers employed in restaurants	1224	9.799	3.429	0	23.045	.275	3.131
Percentage of teenagers employed in retail	1224	19.389	5.685	2.564	38.073	.172	3.048
Employment percentage of females	1224	37.634	11.11	1.242	73.668	.152	3.06
Employment percentage of males	1224	35.403	10.879	0	71.537	.118	2.885
Part-time employment percentage of females	1224	27.704	9.476	0	64.952	.397	3.407
Part-time employment percentage of males	1224	23.1	8.688	0	54.027	.362	2.998
Full-time employment percentage of females	1224	9.931	4.998	0	36.312	.781	4.105
Full-time employment percentage of males	1224	12.303	5.532	0	39.618	.622	3.803
Minimum wage (in US dollars)	1224	5.522	1.281	3.35	9.19	.5	2.215
Real minimum wage	1224	3.057	.307	2.509	4.024	.572	2.963
Logarithm of the real minimum wage	1224	1.112	.099	.92	1.392	.337	2.676
Source: Author's calculations							

Table 5: Matrix of correlations of teenage employment in different sectors

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Employed %	1.000											
Part-time %	0.905	1.000										
Full-time %	0.612	0.218	1.000									
In school %	-0.197	0.084	-0.610	1.000								
Restaurant sector%	0.572	0.559	0.273	-0.027	1.000							
Retail sector %	0.841	0.829	0.389	-0.048	0.729	1.000						
Female %	0.921	0.854	0.525	-0.131	0.558	0.782	1.000					
Male %	0.921	0.811	0.604	-0.231	0.493	0.763	0.698	1.000				
Part-time female %	0.819	0.909	0.189	0.105	0.536	0.753	0.894	0.615	1.000			
Part-time male %	0.818	0.896	0.210	0.046	0.469	0.742	0.644	0.863	0.634	1.000		
Full-time female %	0.494	0.175	0.809	-0.490	0.222	0.311	0.528	0.387	0.092	0.230	1.000	
Full-time male %	0.526	0.188	0.858	-0.525	0.233	0.337	0.362	0.611	0.213	0.128	0.399	1.000

Source: Author's calculations

Methodology

This study combines data from 51 states with employment records for teenagers aged 16 to 19 to estimate the causal impact of state-level minimum wage increases. We build upon the empirical methodology used by Callaway et. al. (2021) and employ a statistical framework for conducting a staggered DID. This allows us to capture variation in the timing of the treatment and group-specific responses. The study also analyses the impact on real wages, providing a macroeconomic comparison of labour market research. The comparison ensures that the identified treatment reflects actual changes in the financial burden faced by employers and workers. The primary causal parameter of interest is the average treatment effect of minimum wage increases on teenage employment, estimated with a staggered DID. While we stagger treatment adoption over time across states, we estimate a pooled post-treatment average effect across all treatment groups and years rather than a per-group time average treatment effect using panel data, which validates the assumption of random variables without strict conditioning on treatments. This slight adjustment is used as we are interested in the policy impact rather than cohort-specific responses, and given the limited sample sizes for treatment cohorts and post-treatment periods, especially for smaller states or states treated late in the sample, separately estimating $ATT(g, t)$ for every group-time combination could lead to increased bias in our aggregate results.

The Staggered DID

Different states experience wage increases independently of one another. This is a general policy response to CPI indexing that reduces reliance on public infrastructure and political initiatives. Therefore, we define the treatment groups as states that have a state-level minimum wage higher than the federal minimum wage, selecting the first year it exceeds the federal minimum to support the assumption of irreversibility of treatment, as without it, we risk contamination of control and treatment groups. Therefore, the treatment indicator for each state s at time t is:

$$D_{st} = 1\{t \geq G_s\} \quad (1)$$

where D_{st} is 1 if the state s is treated at the time t and G_s is the first year when the state minimum wage exceeds the Federal minimum wage. As referred to in Callaway (2021), the cohort year treatment effect is:

$$ATT(g, t) = \mathbb{E}[Y_{st}(1) - Y_{st}(0) \mid G_s = g] \quad (2)$$

where $Y_{st}(1)$ is the teenage employment rate if the state s is treated, and $Y_{st}(0)$ is the control group. Therefore, we are seeing how much treatment affected teen employment in year t . If we estimate $ATT(g, t)$ for each group, it would provide results that show how distinct groups of states responded over time. However, the goal of this study is to see how the minimum wage increases affected teenage employment on average to get an overall picture. Therefore, we still treat years as the staggered model would, but estimate a pooled post-treatment average effect:

$$Y_{st} = \alpha_s + \beta(\text{Treatment}_s \times \text{Post}_t) + D_t' \lambda + \varepsilon_{st} \quad (3)$$

In equation 3, we have considered the following parameters for this study; we have not included the state dummies as the fixed effect panel regression already accounts for state-level differences. Y_{st} is the outcome variable for state s in year t . The outcome variables we have considered in this study are listed in the results table and equations 8-16 (appendix). Treatment is the dummy variable which takes the value of 1 if state s is in the treatment group and 0 if state s is in the control group. Post is the dummy variable that takes the value of 1 if the year t is the first year the state minimum wage is greater than the federal minimum wage, and all subsequent years after t , and it takes the value 0 for all years before the year t . The Treatment $\times$ Post is the interaction term, 1 if treated state s in a post-treatment year t and 0 otherwise. D_t' is a 23×1 vector of dummy variables indicating the year from 1991 to 2013. The base year is 1990, but we have not taken a dummy variable for the base variable in order to avoid the dummy variable trap. Therefore, all results with respect to time fixed effects will be in comparison to the base year 1990. β is the coefficient for (Treatment $\times$ Post), the DID estimate. α_s is the intercept. λ is the vector which consists of estimated coefficients capturing the time fixed effects, it is also a 23×1 vector. Where the base year is 1990, therefore, all results with respect to time fixed effects will be in comparison to the base year 1990. The β gives us an estimated summary of the average effect on employment. This is relevant to the Callaway et. al (2021) methodology because, as shown by Goodman (2021), the β is actually:

$$\beta = \sum_{g,t} w_{g,t} \times ATT(g, t) \quad (4)$$

where $w_{g,t}$ is the weight placed on each $ATT(g, t)$ i.e. how much each cohort contributes to the overall β . These weights depend on the cohort size and the timing of treatment relative to other states. This means the β is a weighted average that can be influenced more heavily by

certain states that have a greater influence on employment outcomes. β does not represent the direct experience of any one group. There is strong heterogeneity; therefore, the treatment effect varies across groups and time periods. To address these concerns, we provide an event study graph, robustness checks, and validity for the staggered DID assumptions. Our approach focuses on the staggered two-way fixed effects regression as it provides the average impact results.

The March CPS data captures the employment outcomes before the implementation of the new federal or state minimum wage policy; therefore, prior to the specified treatment for any given state, teenagers and employers are unaffected by the wage change, limited anticipation assumption:

$$\mathbb{E}[Y_{st}(g) \mid G_s = g] = \mathbb{E}[Y_{st}(0) \mid G_s = g], \quad \text{for all } t < g, \quad (5)$$

The assumption is important as employers may reduce youth employment in anticipation, causing contamination of the treatment groups and introducing bias. The pre-period reflects a partial impact of the policy, showing that the policy had a lower impact on unemployment than it actually did ($Y_{st}(0)$ would be contaminated by treatment effects). We take two precautionary measures to avoid this. First, we identify the states that had a higher state-level minimum wage than the federal minimum wage in 1990 and omit them from the regression data to ensure every treatment has an observable pre-trend period. , for states treated in 1990, $t = g$ occurs in the first period of the dataset; therefore, there are no earlier years $t < g$ to observe pre-treatment outcomes. Without any pre-treatment data, we cannot construct the first-difference term ($Y_{t>g} - Y_{t<g} \mid \text{Treated}$) properly. Thus, for 1990 states, the DID estimator cannot be constructed due to the missing baseline comparison. Secondly, we conduct a placebo test using years before the first treated year for a state, which is 1993. The result shows that the coefficient of the treatment post variable is negligible and highly insignificant at $p = 0.095$, which indicates no meaningful impact on teenage employment. The treatment variable is statistically insignificant and fails to reject the hypothesis that the treatment and control groups were

Table 6: Placebo Test

Percentage Employed	(1) Coefficient	(2) Standard er.	(3) t-value	(4) p-value	[95% Conf.	(5) Interval]	(6) Sig.
Treatment	-3.056	2.747	-1.11	.274	-8.639	2.526	
Post	-6.148	1.584	-3.88	0	-9.366	-2.93	***
Treat x post	.082	1.606	0.05	.959	-3.181	3.346	
1990	0						
1991	-3.932	1.137	-3.46	.001	-6.244	-1.621	***
1992	-7.765	1.073	-7.23	0	-9.946	-5.583	***
1993	0						
Constant	44.469	2.012	22.10	0	40.38	48.559	***

*** p<.01, ** p<.05, * p<.1

statistically different before treatment. The post variable is significant at the 1% level, showing that teenage employment eventually declines, but the decline is attributed to common time trends rather than any anticipatory effects. The estimated effects from the treatment can be interpreted as reflecting genuine responses to the minimum wage policy.

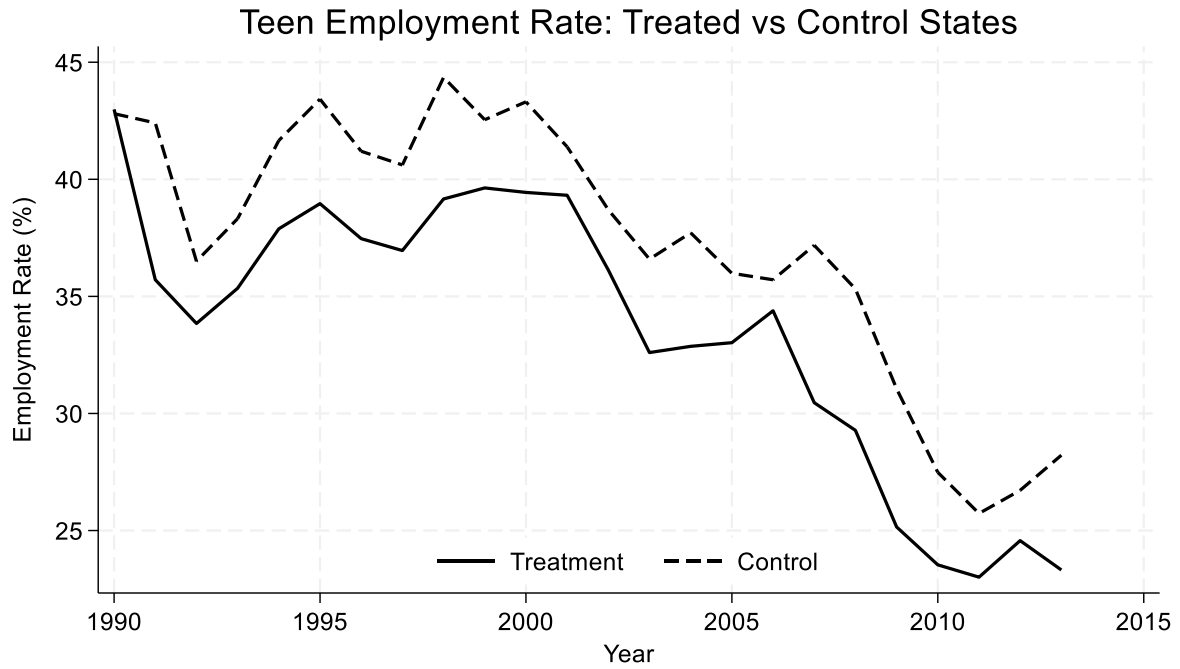
Both treatment and control groups cover a wide range of characteristics of the distribution of employment in different sectors, industries and demographics. The overlap condition requires the baseline covariates or features of the state-level data to be diverse so that no state is deterministically assigned based on its composition, which essentially imposes uncertainty on the probability of a covariate feature of the state being treated or untreated. This ensures that every treated feature has an untreated counterpart. The analysed cohorts also face minimal interference as we do not use border-level comparisons, and the likelihood of teenagers 16-19 commuting out of state for work is highly unlikely; therefore, the treatment of one state does not affect another.

Conditional parallel trends are essential to validate the utilisation of DID, as we assume that in the absence of treatment, the treatment group would behave similarly to the control group:

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) | X, G_s = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) | X, C_s = 1] \quad (6)$$

where $C_s = 1$ denotes the control group. The DID analysis takes this into account by selecting states that have never been treated and eliminating 1990 treatment states that contaminate the counterfactual.

Figure 3: Parallel Trends Graph



From Figure 4, we can observe existing parallel trends between control and treatment groups, as seen until the late 1990s. The composition of treated states was relatively low, suggesting that the treatment groups behaved like controls in the absence of a minimum wage rise, indicating parallel trends and feature similarity. Under the established presence of parallel trends, we can employ

$$\widehat{ATT}(g, t) = (\bar{Y}_{g,t} - \bar{Y}_{g,t-1}) - (\bar{Y}_{C,t} - \bar{Y}_{C,t-1}) \quad (7)$$

The estimated Average Treatment Effect on the Treated (ATT) for the cohort g in year t is calculated using the change in outcomes for the treated group minus the change in outcomes for the control group. Where, $\bar{Y}_{g,t}$ represents the average teenage employment rate for the treated group in the year t , while $\bar{Y}_{g,t-1}$ represents the average teenage employment rate for the same group in the year prior, and the C subset counterparts represent the control groups. The difference gives us the isolated causal effect of policy under the assumption of parallel trends. As teenage jobs are concentrated around low-paying jobs, and minimum wage rises are staggered, meeting the formal conditions allows us to credibly capture the average causal effect of real minimum wage on teenage employment across US states between 1990 and 2013.

Results

Table 7: Results from the panel regression of treatment states on the employment of teenagers in the US based on nominal minimum wages

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables	Employed percentage	Employed percentage	Employment percentage of females	Employment percentage of males	Full-time employment percentage	Part-time employment percentage	Part-time employment percentage of females	Part-time employment percentage of males	Full-time employment percentage of females	Full-time employment percentage of males
Treatment*Post	-1.316** (0.645)	-1.505** (0.641)	-2.068** (0.831)	-0.500 (0.866)	-1.409*** (0.431)	0.092 (0.548)	-0.410 (0.749)	0.751 (0.725)	-1.658*** (0.553)	-1.251** (0.617)
Constant	42.898*** (0.787)	42.898*** (1.379)	44.014*** (1.015)	42.014*** (1.057)	15.268*** (0.526)	27.630*** (0.670)	29.902*** (0.915)	25.434*** (0.886)	14.112*** (0.675)	16.580*** (0.754)
Observations	840	840	840	840	840	840	840	840	840	840
R-squared	0.613	0.613	0.469	0.495	0.390	0.441	0.291	0.334	0.253	0.280
Number of states	35	35	35	35	35	35	35	35	35	35
Sigma u	7.118	6.675	7.551	6.920	2.437	5.913	6.242	5.737	2.545	2.565
Sigma e	4.658	4.658	6.002	6.255	3.112	3.962	5.410	5.240	3.991	4.458
Rho	0.700	0.672	0.613	0.550	0.380	0.690	0.571	0.545	0.289	0.249

*** p<0.01, ** p<0.05, * p<0.1

Notes. Standard errors are shown in parentheses. The variable Post is a dummy equal to 1 for years when the state-level minimum nominal wage was above the federal minimum nominal wage for the first time, and thereafter, all the years after it are also 1, and 0 for years before it. Treatment is a dummy equal to 1 if the state had a minimum nominal wage above the federal minimum nominal wage; otherwise, it is 0. The coefficients on Treatment*Post capture the Difference in Difference estimate of the effect of the state-level minimum nominal wage being above the federal minimum nominal wage on teenage employment.

All columns except Column (2) report fixed effect results. Only Column (2) reports random effect results.

Table 8: Results from the panel regression of treatment states on the employment of teenagers in the US based on the real minimum wages

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables	Employed percentage	Employed percentage	Employment percentage of females	Employment percentage of males	Full-time employment percentage	Part-time employment percentage	Part-time employment percentage of females	Part-time employment percentage of males	Full-time employment percentage of females	Full-time employment percentage of males
Treatment*Post	-1.320** (0.563)	-1.406** (0.558)	-1.514** (0.733)	-1.200 (0.760)	-0.736** (0.375)	-0.584 (0.481)	-0.684 (0.668)	-0.521 (0.653)	-0.830* (0.481)	-0.679 (0.532)
Constant	43.438*** (0.696)	43.438*** (1.273)	44.078*** (0.906)	43.024*** (0.940)	14.599*** (0.464)	28.839*** (0.595)	30.713*** (0.826)	27.034*** (0.808)	13.364*** (0.595)	15.990*** (0.658)
Observations	1,104	1,104	1,104	1,104	1,104	1,104	1,104	1,104	1,104	1,104
R-squared	0.587	0.587	0.429	0.473	0.345	0.424	0.268	0.302	0.213	0.250
Number of states	46	46	46	46	46	46	46	46	46	46
Sigma u	7.321	7.225	7.978	6.914	2.306	6.565	7.241	6.071	2.409	2.442
Sigma e	4.723	4.723	6.147	6.374	3.146	4.034	5.601	5.477	4.038	4.460
Rho	0.706	0.701	0.627	0.541	0.349	0.726	0.626	0.551	0.263	0.231

*** p<0.01, ** p<0.05, * p<0.1

Notes. Standard errors are shown in parentheses. The variable Post is a dummy equal to 1 for years when the state-level minimum real wage was above the federal minimum real wage for the first time, and thereafter, all the years after it are also 1, and 0 for years before it. Treatment is a dummy equal to 1 if the state had a minimum real wage above the federal minimum real wage; otherwise, it is 0. The coefficients on Treatment*Post capture the Difference in Difference estimate of the effect of the state-level minimum real wage being above the federal minimum real wage on teenage employment.

All columns except Column (2) report fixed effect results. Only Column (2) reports random effect results.

Findings and Policy Implications

The estimates show that state minimum wage increases reduce teenage employment by 1.316 percentage points on average when using nominal wages and a reduction in teenage employment by 1.320 percentage points on average when using real wage differences. Both results are significant at the 5% level, indicating we reject the null hypothesis of no effect on employment. Our model explains 61.7% and 58.7% of the variation in teenage unemployment for nominal and real wage differences, respectively. The similarity between nominal and real wages suggests that inflation adjustment does not substantially affect teenage employment. However, as evidenced in the literature, nominal wage shocks can have a slightly greater impact in the short run as firms respond to nominal cost increases and adjust to inflation-adjusted conditions over time (O'Hanlon & Peasnell, 2002). Constant terms indicate the base level of the teenage employment rate to be 42.89% and 43.83% for nominal and real wages, respectively, holding other factors constant. An important comparison is the 3.7% negative impact on teenage employment from the Callaway (2021) findings; the slightly lower impact can be attributed to the treatment and control group composition and a much smaller window of analysis, from 2001 to 2007. Therefore, their study provides a strong identification of short-run dynamic effects. In contrast, using data from 1990 to 2013, we find a lower impact on employment, likely due to state composition heterogeneity estimated in the treatment effects. Despite the macroeconomic differences in the data used, the magnitude of our findings remains comparable to the original paper.

In both of our models, the consistently high Rho values indicate intra-state feature correlation and less random noise; therefore, the substantial proportion of residual variance in teenage employment is contributed to time-invariant state-level characteristics rather than time trends. This validates the use of fixed effects estimators, which rely on the state characteristics changes over time, making our results coordinated with policy changes. The standard deviation of unobserved state-level effects (Sigma u) consistently exceeds the standard deviation of the error component (Sigma e) across all models, suggesting cross-sectional heterogeneity across states is more dominant than random shocks. This is validated by data collection on state-level minimum wage policy to reflect a larger preexisting economy, like in California.

Policy analysis of these findings should exercise great caution when interpreting these results. Due to the lack of weighted cohort analysis or synthetic controls, the negative impacts partially reflect unobserved biases and macroeconomic phenomena unrelated to the direct impact on teenage unemployment. The existing literature provides strong evidence for labour substitution

and methodological restrictions in the analysis of minimum wages. However, if we consider only the average causal effect, the most viable policy implementations are the UK's systemic minimum wage for younger individuals, which encourages employment, and the introduction of youth-focused training programs for out-of-school individuals. The analysis of case studies from Europe suggests divisional unions that specialise in the renewal of youth labour using skill training programs (Tapia & Turner, 2018).

The Gender cohort estimates show that females experience substantially larger employment losses than men. We find that female teenagers aged 16-19 face a fall in employment by 2.068 percentage points on average when using differences in nominal wages, and a 1.514 percentage points decrease using real wages. Both estimates are significant at the 5% level. We find the average impact of a minimum wage rise on teenage male employment to be insignificant. Therefore, we cannot reject the null hypothesis that the policy has no effect in both nominal and inflation-adjusted circumstances. The fixed effects model shows moderate explanatory power for these findings, 46.9% for females using nominal wage, and 49.5% for males, real wages maintain similar explanatory power. The possible explanations for these results include the disproportionate inclusion of women in low-wage sectors such as retail and hospitality, sectors that are highly sensitive to minimum wage changes (Neumark & Wascher, 2008). Additionally, women are more likely than men to leave the labour market search entirely, especially at a young age, due to educational commitments (Leana & Feldman, 1991). These findings account for a small fraction of the supply side of the labour force; therefore, effective policy implementation to encourage hiring is essential.

The introduction of Active Labour Market Programs introduces skills training programs, helping teenagers gain sufficient technical skills. Entrepreneurship support for young people starting their own business. Job search assistance with incentivised help in CV writing and interview preparations. Subsidised wages to encourage firms to hire young workers (Kluve et al., 2016). Research finds strong long-term effects on employment and earnings after the implementation of these programs (Kuka et al, 2020). Youth employment is most heavily impacted by economic downturn, expansive fiscal policies and targeted investment in sectors with potential youth jobs can counteract this significantly (*International Labour Organisation, 2017*).

Lastly, the estimates for teenage employment contracts show a significant impact on full-time jobs. Full-time employment rates on average decline by 1.409 percentage points for nominal differences and 0.736 percentage points for the real wage. Statistically significant at the 1%

and 5% level, respectively. The model explains 61.3% and 58.7% of the variation in job loss for nominal and real, respectively. In comparison, there is no statistically significant change in the level of employment for part-time workers in either of our models. The effect is even more prominent among females in full-time employment, where estimates show that on average, a minimum wage rise causes a 1.658 percentage point fall in female employment using nominal wage and a 0.830 percentage point fall using real wage, where results are significant at the 1% and 10% level, respectively.

Results strongly support the dynamic monopsony model, due to the initial nominal wage rises, employers are likely to hire more skilled labour as they anticipate and increase production. The negative impact in the real wage analysis suggests a lower demand for low-margin goods like fast food and retail, where employers are likely to reduce full-time slots, which represent larger unit costs. The difference in unemployment is also reflected in the findings of McDonald et al.(2009), which finds that the part-time jobs labour flows face less redundancy through a compromise in real wages or no non-wage benefits.

These contractions highlight the need for policies aimed at preserving stable youth job opportunities. Policymakers should target subsidies for firms hiring teenagers into full-time roles, expanding apprenticeship pathways for full-time employment. The most difficult transition is from part-time to full-time, government initiatives and programs that target low-margin sectors and implement policy to facilitate minimum wage rises, while addressing the adverse macroeconomic impacts on youth employment.

Conclusion

This study finds preliminary evidence of unemployment from minimum wage rises. We address the assumptions for a DID analysis and find average negative treatment effects on overall employment and cohort-specific job losses. Although we get statistically significant results following the literature and statistical implications, the direct casualty should be interpreted with caution.

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Appendix

DID Equations:

$$employedpercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (8)$$

$$employedfpercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (9)$$

$$employedmpercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (10)$$

$$workfulltimepercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (11)$$

$$workparttimepercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (12)$$

$$workparttimefemalepercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (13)$$

$$workparttimemalepercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (14)$$

$$workfulltimefemalepercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (15)$$

$$workfulltimemalepercent_{st} = \alpha_s + \beta(Treatment_s \times Post_t) + D'_t\lambda + \epsilon_{st} \quad (16)$$

Table 9: Results from the panel regression of treatment states on the employment of teenagers in the US based on nominal minimum wage

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables	Employed percentage	Employed percentage	Employment percentage of females	Employment percentage of males	Full-time employment percentage	Part-time employment percentage	Part-time employment percentage of females	Part-time employment percentage of males	Full-time employment percentage of females	Full-time employment percentage of males
Treatment*Post	-1.316** (0.645)	-1.505** (0.641)	-2.068** (0.831)	-0.500 (0.866)	-1.409*** (0.431)	0.092 (0.548)	-0.410 (0.749)	0.751 (0.725)	-1.658*** (0.553)	-1.251** (0.617)
Year-fixed effects(Base Year = 1990)										
1991	-3.932*** (1.114)	-3.932*** (1.116)	-4.002*** (1.435)	-3.774** (1.495)	-2.105*** (0.744)	-1.827* (0.947)	-1.672 (1.293)	-1.945 (1.253)	-2.329** (0.954)	-1.829* (1.066)
1992	-7.765*** (1.114)	-7.765*** (1.116)	-8.396*** (1.435)	-7.477*** (1.495)	-3.646*** (0.744)	-4.119*** (0.947)	-4.801*** (1.293)	-3.592*** (1.253)	-3.595*** (0.954)	-3.884*** (1.066)
1993	-6.068*** (1.114)	-6.062*** (1.116)	-8.651*** (1.435)	-3.771** (1.495)	-4.358*** (0.744)	-1.710* (0.947)	-4.273*** (1.293)	0.886 (1.253)	-4.378*** (0.954)	-4.657*** (1.066)
1994	-3.104*** (1.114)	-3.093*** (1.117)	-4.902*** (1.436)	-1.657 (1.496)	-3.515*** (0.744)	0.411 (0.948)	-1.135 (1.294)	1.731 (1.253)	-3.767*** (0.955)	-3.388*** (1.066)
1995	-1.692 (1.114)	-1.681 (1.117)	-2.886** (1.436)	-0.560 (1.496)	-1.556** (0.744)	-0.136 (0.948)	-0.519 (1.294)	0.268 (1.253)	-2.368** (0.955)	-0.827 (1.066)
1996	-3.543*** (1.114)	-3.533*** (1.117)	-4.807*** (1.436)	-2.859* (1.496)	-1.818** (0.744)	-1.725* (0.948)	-1.399 (1.294)	-2.401* (1.253)	-3.408*** (0.955)	-0.458 (1.066)
1997	-4.050*** (1.115)	-4.034*** (1.117)	-3.526** (1.437)	-4.788*** (1.497)	-1.700** (0.745)	-2.350** (0.948)	-2.588** (1.295)	-2.076* (1.254)	-0.938 (0.955)	-2.712** (1.067)
1998	-1.101 (1.115)	-1.085 (1.117)	-0.292 (1.437)	-2.068 (1.497)	-0.835 (0.745)	-0.266 (0.948)	0.327 (1.295)	-0.860 (1.254)	-0.619 (0.955)	-1.208 (1.067)
1999	-1.737 (1.115)	-1.721 (1.117)	-2.067 (1.437)	-1.744 (1.497)	-1.564** (0.745)	-0.173 (0.948)	-0.397 (1.295)	-0.045 (1.254)	-1.671* (0.955)	-1.699 (1.067)
2000	-1.465 (1.115)	-1.448 (1.117)	-3.093** (1.437)	-0.156 (1.497)	-0.893 (0.745)	-0.572 (0.948)	-1.295 (1.295)	0.166 (1.254)	-1.797* (0.955)	-0.322 (1.067)
2001	-2.461** (1.115)	-2.445** (1.117)	-2.500* (1.437)	-2.552* (1.497)	-0.784 (0.745)	-1.676* (0.948)	-1.877 (1.295)	-1.480 (1.254)	-0.623 (0.955)	-1.071 (1.067)
2002	-5.408*** (1.115)	-5.392*** (1.117)	-6.080*** (1.437)	-5.069*** (1.497)	-2.231*** (0.745)	-3.176*** (0.948)	-3.112** (1.295)	-3.417*** (1.254)	-2.968*** (0.955)	-1.652 (1.067)
2003	-8.245*** (1.115)	-8.229*** (1.117)	-7.670*** (1.437)	-9.060*** (1.497)	-4.225*** (0.745)	-4.020*** (0.948)	-3.802*** (1.295)	-4.330*** (1.254)	-3.869*** (0.955)	-4.730*** (1.067)

	Dependent variable	Constant	Age	Age ²	Female	Married	White	Black	Hispanic	Other race
2004	-7.525***	-7.503***	-8.334***	-6.867***	-4.032***	-3.492***	-4.022***	-3.088**	-4.312***	-3.779***
	(1.116)	(1.119)	(1.438)	(1.498)	(0.746)	(0.949)	(1.296)	(1.255)	(0.956)	(1.068)
2005	-8.247***	-8.220***	-8.158***	-8.552***	-4.122***	-4.125***	-4.269***	-4.013***	-3.889***	-4.539***
	(1.117)	(1.120)	(1.440)	(1.500)	(0.746)	(0.950)	(1.298)	(1.257)	(0.957)	(1.069)
2006	-7.642***	-7.610***	-8.279***	-7.326***	-3.216***	-4.426***	-3.968***	-4.931***	-4.311***	-2.395**
	(1.119)	(1.122)	(1.442)	(1.503)	(0.748)	(0.952)	(1.300)	(1.259)	(0.959)	(1.071)
2007	-8.532***	-8.440***	-8.502***	-8.891***	-3.272***	-5.260***	-4.857***	-5.853***	-3.645***	-3.039***
	(1.157)	(1.159)	(1.490)	(1.553)	(0.773)	(0.984)	(1.344)	(1.301)	(0.991)	(1.107)
2008	-10.007***	-9.910***	-9.723***	-10.465***	-4.215***	-5.792***	-5.445***	-6.296***	-4.278***	-4.169***
	(1.162)	(1.164)	(1.497)	(1.560)	(0.776)	(0.988)	(1.349)	(1.307)	(0.996)	(1.112)
2009	-14.202***	-14.105***	-13.762***	-14.947***	-6.121***	-8.080***	-7.858***	-8.498***	-5.904***	-6.449***
	(1.162)	(1.164)	(1.497)	(1.560)	(0.776)	(0.988)	(1.349)	(1.307)	(0.996)	(1.112)
2010	-16.766***	-16.669***	-16.203***	-17.304***	-7.342***	-9.424***	-9.521***	-9.182***	-6.683***	-8.122***
	(1.162)	(1.164)	(1.497)	(1.560)	(0.776)	(0.988)	(1.349)	(1.307)	(0.996)	(1.112)
2011	-17.889***	-17.792***	-17.817***	-18.288***	-8.070***	-9.819***	-10.072***	-9.767***	-7.744***	-8.521***
	(1.162)	(1.164)	(1.497)	(1.560)	(0.776)	(0.988)	(1.349)	(1.307)	(0.996)	(1.112)
2012	-16.607***	-16.510***	-16.044***	-17.546***	-6.229***	-10.379***	-10.693***	-10.247***	-5.351***	-7.299***
	(1.162)	(1.164)	(1.497)	(1.560)	(0.776)	(0.988)	(1.349)	(1.307)	(0.996)	(1.112)
2013	-16.523***	-16.426***	-15.482***	-17.829***	-6.265***	-10.258***	-9.531***	-11.116***	-5.952***	-6.714***
	(1.162)	(1.164)	(1.497)	(1.560)	(0.776)	(0.988)	(1.349)	(1.307)	(0.996)	(1.112)
Constant	42.898***	42.898***	44.014***	42.014***	15.268***	27.630***	29.902***	25.434***	14.112***	16.580***
	(0.787)	(1.379)	(1.015)	(1.057)	(0.526)	(0.670)	(0.915)	(0.886)	(0.675)	(0.754)
Observations	840	840	840	840	840	840	840	840	840	840
R-squared	0.613	0.613	0.469	0.495	0.39	0.441	0.291	0.334	0.253	0.28
Numberof states	35	35	35	35	35	35	35	35	35	35
Sigma u	7.118	6.675	7.551	6.92	2.437	5.913	6.242	5.737	2.545	2.565
Sigma e	4.658	4.658	6.002	6.255	3.112	3.962	5.41	5.24	3.991	4.458
Rho	0.7	0.672	0.613	0.55	0.38	0.69	0.571	0.545	0.289	0.249

*** p<0.01, ** p<0.05, * p<0.1 Notes. Standard errors are shown in parentheses. The variable Post is a dummy equal to 1 for years when the state-level minimum nominal wage was above the federal minimum nominal wage for the first time, and thereafter, all the years after it are also 1, and 0 for years before it. Treatment is a dummy equal to 1 if the state had a minimum nominal wage above the federal minimum nominal wage; otherwise, it is 0. The coefficients on Treatment*Post capture the Difference in Difference estimate of the effect of the state-level minimum nominal wage being above the federal minimum nominal wage on teenage employment. All columns except Column (2) report fixed effect results. Only Column (2) reports random effect results.

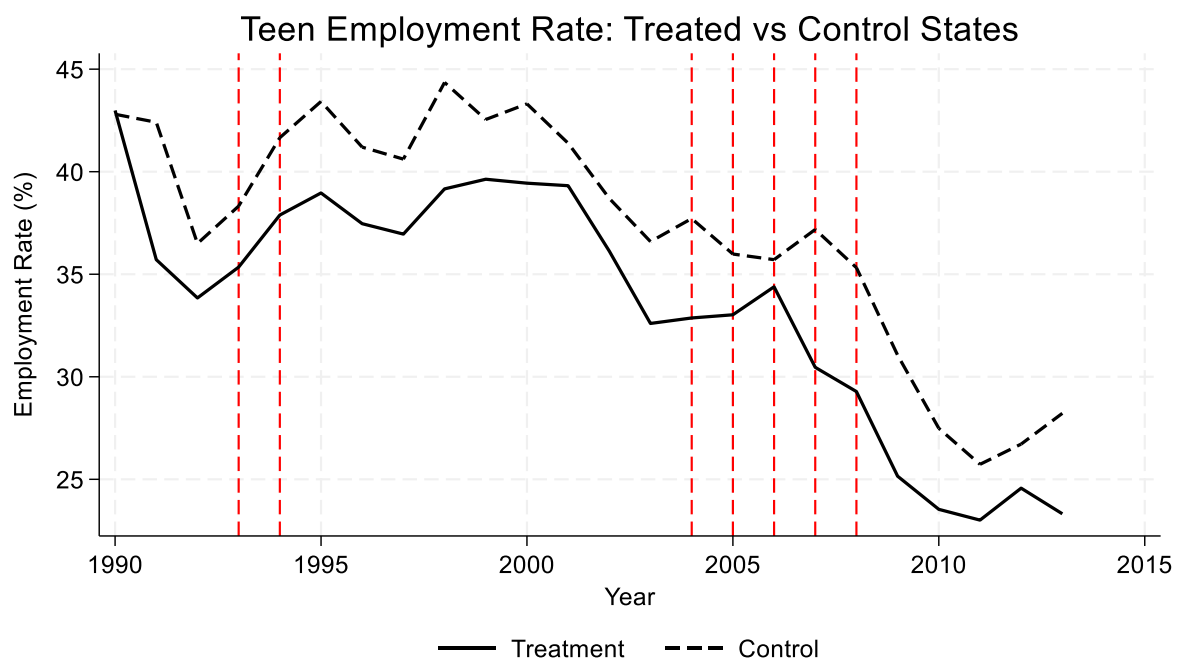
Table 10: Results from the panel regression of treatment states on the employment of teenagers in the US based on the real minimum wage

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Variables	Employed percentage	Employed percentage	Employment percentage of females	Employment percentage of males	Full-time employment percentage	Part-time employment percentage	Part-time employment percentage of females	Part-time employment percentage of males	Full-time employment percentage of females	Full-time employment percentage of males
Treatment*Post	-1.320** (0.563)	-1.406** (0.558)	-1.514** (0.733)	-1.200 (0.760)	-0.736** (0.375)	-0.584 (0.481)	-0.684 (0.668)	-0.521 (0.653)	-0.830* (0.481)	-0.679 (0.532)
Year-fixed effects(Base Year=1990)										
1991	-3.159*** (0.985)	-3.159*** (0.985)	-2.800** (1.282)	-3.464*** (1.329)	-1.497** (0.656)	-1.663** (0.841)	-1.041 (1.168)	-2.115* (1.142)	-1.759** (0.842)	-1.349 (0.930)
1992	-7.069*** (0.985)	-7.069*** (0.985)	-6.403*** (1.282)	-7.868*** (1.329)	-3.577*** (0.656)	-3.492*** (0.841)	-3.342*** (1.168)	-3.678*** (1.142)	-3.061*** (0.842)	-4.190*** (0.930)
1993	-5.663*** (0.985)	-5.660*** (0.985)	-6.792*** (1.282)	-4.474*** (1.329)	-3.509*** (0.656)	-2.154** (0.841)	-2.979** (1.168)	-1.080 (1.142)	-3.813*** (0.842)	-3.394*** (0.930)
1994	-3.328*** (0.985)	-3.324*** (0.985)	-3.664*** (1.282)	-3.238** (1.329)	-3.058*** (0.656)	-0.270 (0.841)	-0.717 (1.168)	0.093 (1.142)	-2.947*** (0.842)	-3.331*** (0.930)
1995	-0.997 (0.986)	-0.991 (0.986)	-1.173 (1.283)	-0.813 (1.330)	-1.436** (0.656)	0.439 (0.842)	1.080 (1.169)	-0.190 (1.143)	-2.253*** (0.843)	-0.623 (0.931)
1996	-3.002*** (0.986)	-2.996*** (0.986)	-3.642*** (1.283)	-2.820** (1.330)	-1.811*** (0.656)	-1.191 (0.842)	-0.562 (1.169)	-2.001* (1.143)	-3.080*** (0.843)	-0.820 (0.931)
1997	-3.005*** (0.986)	-3.000*** (0.986)	-1.623 (1.283)	-4.580*** (1.330)	-1.148* (0.656)	-1.857** (0.842)	-1.009 (1.169)	-2.641** (1.143)	-0.614 (0.843)	-1.939** (0.931)
1998	-0.739 (0.986)	-0.734 (0.986)	0.545 (1.283)	-2.346* (1.330)	-1.142* (0.656)	0.403 (0.842)	1.518 (1.169)	-0.823 (1.143)	-0.974 (0.843)	-1.523 (0.931)
1999	-1.170 (0.986)	-1.165 (0.986)	-0.958 (1.283)	-1.652 (1.330)	-1.579** (0.656)	0.409 (0.842)	1.061 (1.169)	-0.273 (1.143)	-2.019** (0.843)	-1.379 (0.931)
2000	-0.648 (0.986)	-0.643 (0.986)	-1.862 (1.283)	0.215 (1.330)	-1.000 (0.656)	0.352 (0.842)	-0.055 (1.169)	0.677 (1.143)	-1.807** (0.843)	-0.462 (0.931)
2001	-1.251 (0.988)	-1.240 (0.988)	-0.680 (1.285)	-1.857 (1.333)	-0.687 (0.658)	-0.564 (0.844)	-0.176 (1.171)	-0.912 (1.145)	-0.504 (0.844)	-0.944 (0.933)
2002	-4.471*** (0.988)	-4.460*** (0.988)	-4.131*** (1.285)	-5.045*** (1.333)	-1.627** (0.658)	-2.844*** (0.844)	-2.341** (1.171)	-3.380*** (1.145)	-1.790** (0.844)	-1.665* (0.933)
2003	-7.221*** (0.990)	-7.206*** (0.990)	-5.915*** (1.288)	-8.751*** (1.336)	-4.212*** (0.659)	-3.009*** (0.845)	-2.208* (1.174)	-3.877*** (1.148)	-3.707*** (0.846)	-4.874*** (0.935)
2004	-6.972*** (0.990)	-6.957*** (0.990)	-6.855*** (1.288)	-7.251*** (1.336)	-3.746*** (0.659)	-3.226*** (0.845)	-2.950** (1.174)	-3.564*** (1.148)	-3.905*** (0.846)	-3.687*** (0.935)
2005	-7.049*** (0.991)	-7.032*** (0.991)	-6.188*** (1.290)	-8.040*** (1.337)	-3.649*** (0.660)	-3.399*** (0.846)	-2.835** (1.175)	-3.882*** (1.149)	-3.353*** (0.847)	-4.158*** (0.936)

2006	-7.054*** (0.994)	-7.033*** (0.994)	-6.833*** (1.294)	-7.443*** (1.341)	-3.291*** (0.662)	-3.763*** (0.849)	-3.037** (1.179)	-4.440*** (1.153)	-3.796*** (0.850)	-3.003*** (0.939)
2007	-7.572*** (1.015)	-7.535*** (1.015)	-7.021*** (1.321)	-8.373*** (1.369)	-3.204*** (0.676)	-4.367*** (0.867)	-3.581*** (1.203)	-5.237*** (1.177)	-3.440*** (0.868)	-3.135*** (0.958)
2008	-8.666*** (1.018)	-8.627*** (1.017)	-7.788*** (1.325)	-9.631*** (1.374)	-3.747*** (0.678)	-4.918*** (0.869)	-4.043*** (1.207)	-5.779*** (1.180)	-3.745*** (0.870)	-3.852*** (0.961)
2009	-13.249*** (1.018)	-13.210*** (1.017)	-12.235*** (1.325)	-14.461*** (1.374)	-6.144*** (0.678)	-7.105*** (0.869)	-6.128*** (1.207)	-8.099*** (1.180)	-6.107*** (0.870)	-6.362*** (0.961)
2010	-15.907*** (1.018)	-15.868*** (1.017)	-14.709*** (1.325)	-16.986*** (1.374)	-7.159*** (0.678)	-8.748*** (0.869)	-8.340*** (1.207)	-8.893*** (1.180)	-6.368*** (0.870)	-8.092*** (0.961)
2011	-16.692*** (1.018)	-16.653*** (1.017)	-15.782*** (1.325)	-17.866*** (1.374)	-7.742*** (0.678)	-8.950*** (0.869)	-8.172*** (1.207)	-9.790*** (1.180)	-7.610*** (0.870)	-8.077*** (0.961)
2012	-15.729*** (1.018)	-15.690*** (1.017)	-15.100*** (1.325)	-16.677*** (1.374)	-6.093*** (0.678)	-9.636*** (0.869)	-9.742*** (1.207)	-9.603*** (1.180)	-5.358*** (0.870)	-7.074*** (0.961)
2013	-15.583*** (1.018)	-15.544*** (1.017)	-13.950*** (1.325)	-17.382*** (1.374)	-5.860*** (0.678)	-9.722*** (0.869)	-8.536*** (1.207)	-10.915*** (1.180)	-5.414*** (0.870)	-6.467*** (0.961)
Constant	43.438*** (0.696)	43.438*** (1.273)	44.078*** (0.906)	43.024*** (0.940)	14.599*** (0.464)	28.839*** (0.595)	30.713*** (0.826)	27.034*** (0.808)	13.364*** (0.595)	15.990*** (0.658)
Observations	1,104	1,104	1,104	1,104	1,104	1,104	1,104	1,104	1,104	1,104
R-squared	0.587	0.587	0.429	0.473	0.345	0.424	0.268	0.302	0.213	0.250
Numberof states	46	46	46	46	46	46	46	46	46	46
Sigma u	7.321	7.225	7.978	6.914	2.306	6.565	7.241	6.071	2.409	2.442
Sigma e	4.723	4.723	6.147	6.374	3.146	4.034	5.601	5.477	4.038	4.460
Rho	0.706	0.701	0.627	0.541	0.349	0.726	0.626	0.551	0.263	0.231

*** p<0.01, ** p<0.05, * p<0.1 Notes. Standard errors are shown in parentheses. The variable Post is a dummy equal to 1 for years when the state-level minimum real wage was above the federal minimum real wage for the first time, and thereafter, all the years after it are also 1, and 0 for years before it. Treatment is a dummy equal to 1 if the state had a minimum real wage above the federal minimum real wage; otherwise, it is 0. The coefficients on Treatment*Post capture the Difference in Difference estimate of the effect of the state-level minimum real wage being above the federal minimum real wage on teenage employment. All columns except Column (2) report fixed effect results. Only Column (2) reports random effect results.

Figure 4: Years of Treatment



STATA DO file included with submission.

END.